



UNDERGROUND COAL MINING (MNC 208)

PROF R M BHATTACHARJEE

Subject	Lecture hrs
Origin of coal, Coalification and factors affecting coalification process, Factors affecting coalification, Theories of origin of coal, Coal resource and their geographical distributions in India	1
Opening of coal seams: Types of mine entries (shaft, incline, adit), Relative advantages and disadvantages, Location of entries	1
Overview of Shaft Sinking Operations: Ground breaking and muck disposal tools and equipment, lining; ventilation, sinking in difficult and water-bearing ground; An overview of mechanized shaft sinking	4
Choice of methods of mining coal seams; Factors affecting choice of mining methods, Inseam mining & Horizon mining	1
Pillar mining methods: a) Layout of B&P / R&P mine, sequence of pillar extraction; b) Formation and extraction of pillars by B&P method using LHDs / SDLs a) Formation and extraction of pillars by R&P method using Continuous Miner - Split and Fender method, Wongawilli method, Rib pillar extraction method b) Statutory provisions regarding formation and extraction of pillars by B&P and R&P method c) Air blast and measures to minimize its effects d) Case studies of R&P mining in India	12
Longwall method of mining: a) Factors affecting longwall mining; advancing and retreating faces; longwall face layouts, factors affecting length and width of longwall panel b) Gate road development using road headers, continuous miner, single entry and multiple entry c) Extraction of coal by longwall method, Longwall face equipment, Application of Shearer / Plough, Armoured Face Conveyor d) Support system of longwall face and gate roads e) Statutory provisions regarding longwall mining f) Case studies longwall faces in India	17
Pillar extraction by stowing; Conditions requiring stowing in mines; types of stowing; suitable materials for hydraulic stowing	2
Introduction to Advanced Technologies	1

Instructions

- ▶ Punctuality and Discipline in classrooms can not be compromised
- ▶ Attendance requirement (75% mandatory)
- ▶ Your attendance will be verified anytime during online classes. If found non-responding, absence will be marked
- ▶ Attendance will be uploaded every week – don't fall short of attendances
- ▶ Read books and journals – lots of papers are available in websites – always update yourself
- ▶ Prepare your own notes – Draw sketch, label it as it is explained
- ▶ Your sessional marks, other than mid-sem marks, will be based on your class notes, assignments and quizzes.

Instructions

- ▶ If you are late by more than 2 minutes, you will not get your attendance on that day
- ▶ If you have any question, please ask
- ▶ You can not expect full marks if you only reproduce the bullet points of the slides. You need to explain each point in your own language, as explained and discussed in the class.

Origin of coal - Coalification

- ▶ Coal – a combustible sedimentary rock formed from accumulation of plant remains modified by chemical, biological and physical processes during and after burial
- ▶ Coal – Originated from vegetable matter – converted to Peat, Lignite, Bituminous coal and Anthracites
- ▶ Coalification is transformation of plant debris under reducing conditions into a continuous series of products having increasing % age of carbon and diminishing % age of oxygen and hydrogen

Origin of coal - Coalification

- ▶ It all starts with a swamp on the edge of a sedimentary basin, such as a lagoon or a lake.
- ▶ Tectonic activity raises sea levels, covering and killing vegetation.
- ▶ Plant debris accumulates and is buried under layers of mud and sand in a process known as sedimentation.
- ▶ This protects the debris from the air and slows down the decomposition process.
- ▶ The vegetation grows back, until the next flooding.

Origin of coal - Coalification

- ▶ The sedimentary basin gradually sinks under the weight of the sediments,
- ▶ Layers of dead plants are subjected to rising temperatures that gradually “cook” them, leading to their transformation.
- ▶ Different stages of sedimentation **turn cellulose**, the main component of wood, from peat to lignite (brown coal), then sub-bituminous coal, followed by bituminous coal and, finally, Anthracite.
- ▶ Anthracite has the highest carbon content

Theories of origin of coal

- ▶ In-situ theory
- ▶ Drift theory

Theories of origin of coal

In-situ theory

- ▶ According to this theory, coal seams are observed where once forest grew.
- ▶ As the land was sinking slowly, the accumulated vegetation matters went under water slowly and did not decompose and destroyed.
- ▶ In the course of time, the rate of sinking of land was increased and coal forest was submerged under water.
- ▶ Again, land along with the coal forest emerged out of water after sufficient time and this cycle went on again and again, which is responsible of formation of coal strata and seams.
- ▶ The evidence of this phenomenon is observed in the coal seam that
 - ▶ the stem of fossil trees is found standing erect with their roots protruding into the underclays.
 - ▶ Consistency in composition and quality of coal seams over wide region suggest the deposition of vegetal matter in still water
- ▶ The uniformity in thickness and composition of coal seams over wide areas suggests that the deposition of plant material took place in still water.

Theories of origin of coal

Drift theory-

This theory tells that, the plant material was transported with the stream of water from one place to another, and finally get deposited in a place of swamp having suitable condition like sediments.

Coal seams of India are of drift origin.

The evidence of drift theory that the rocks associated with the coal seams are sedimentary.

The coal seam itself behaves like a sedimentary bed and they are observed to branch out.

Geological Time for Formation of Coal

- ▶ Most favorable conditions for formation of coal occurred 360 million to 290 million years ago, during the Carboniferous (“coal-bearing”) Period.
- ▶ However, lesser amounts continued to form in some parts of the Earth during all subsequent periods, in particular the Permian (290 million to 250 million years ago), and throughout the Mesozoic Era (250 million to 65 million years ago).
- ▶ The accumulated plant matter buried during the Tertiary Era — less than 65 million years ago — is generally less mature.
- ▶ It is often in the form of lignite, which still contains a high content of volatile matter (bitumen and decayed wood) and has a lower carbon content.
- ▶ However, there is also some higher rank coal from the Tertiary Era, coal that matured early, heated by plate tectonics.
- ▶ Examples of this include Paleocene coal (65 to 55 million years ago), found in Columbia and Venezuela, and Miocene coal (20 million years ago), found in Indonesia.
- ▶ In Indonesia, where the geothermal gradient is very high, anthracite lies close to the surface.

Different Types of Coal

- ▶ Different types of coal are ranked according to their carbon and volatile matter content.
- ▶ Anthracite is 86 to 98% pure carbon and 8 to 3% volatile matter.
- ▶ Bituminous coal contains 70 to 86% carbon and 46 to 31% volatile matter.
- ▶ Sub-bituminous coal is 70 to 76% carbon and 53 to 42% volatile matter.
- ▶ Lignite is 65 to 70% carbon and 63 to 53% volatile matter.
- ▶ Peat consists of partially decomposed vegetation. Technically speaking, it isn't coal. It has a carbon content of less than 60% and is composed entirely of volatile matter.

Coal resource and their geographical distributions in India

- ▶ 319.02 Billion tonnes of Geological Resources of Coal have so far been estimated in the country as on 1.4.2018

State	Proved	Indicated	Inferred	Total
Total	148787	139164	31069	319020
JHARKHAND	45563	31439	6150	83152
ODISHA	37391	34165	7739	79295
CHHATTISGARH	20428	34576	2202	57206
WEST BENGAL	14156	12869	4643	31667
MADHYA PRADESH	11958	12154	3875	27987
TELANGANA	10475	8576	2651	21702
MAHARASHTRA	7178	3074	2048	12299
ANDHRA PRADESH	0	1149	432	1581
BIHAR	161	813	392	1367
UTTAR PRADESH	884	178	0	1062
MEGHALAYA	89	17	471	576
ASSAM	465	57	3	525
NAGALAND	9	0	402	410
SIKKIM	0	58	43	101
ARUNACHAL PRADESH	31	40	19	90

CATEGORIZATION OF RESOURCES

Formation	Proved/Measured	Indicated	Inferred	Total
Grand Total	148787	139164	31069	319020
Gondwana Coals	148194	139065	30174	317433
Tertiary Coals	594	99	895	1588

Type of Coal	Proved/Measured	Indicated	Inferred	Total
Grand Total	148787	139164	31069	319020
(A) Coking :-				
-Prime Coking	4649	664	0	5313
-Medium Coking	13914	11709	1879	27502
-Semi-Coking	519	995	193	1708
Sub-Total Coking	19082	13368	2073	34522
(B) Non-Coking :-	129112	125697	28102	282910
(C) Tertiary Coal :-	594	99	895	1588

Coalfields in India

▶ RANIGANJ COALFIELD

In west Bengal Raniganj coalfield is the important coalfield. This coalfield is the easternmost of the Damodar Valley Coalfields. The coalfields lie mostly in the West Bengal and partly (western portion) in Bihar .

- ▶ Coals of the Barkar Measures are low in moisture (1-3%), low in volatile matter (20-30%) whereas coals of Raniganj Measures are high in moisture (3-10%) and high in volatile matter (30-36%).
- ▶ Other coalfields found in West Bengal are Barjora coalfield, Birbhum coalfield and Darjeeling coalfield.

Coalfields in India

1. Damodar Valley Coalfields:

1. JHARIA COALFIELD

Jharia coalfield is the most important coalfield in India .

This is because this coalfield is now the sole is the prime depository of prime coking coal.

This coalfield is situated about 260 km northwest of Calcutta in the heart of Damodar Valley mainly along the north of this river.

The coalfield lies within the district of Dhanbad. The field is roughly sickle shaped.

The coal basin extends for about 38 km in an east-west direction and a maximum of 18 km in north-south direction and covers an area of about 456 sq. km.

2. BOKARO COALFIELD

- East Bokaro coalfield: There are several coal seams with in the Barkar Formation. Lower coal seams are grouped as Karo Group in which there are 11 coal seams. Out of these seams
- Karo Lower Measures (19-64m) and Karo Upper Measures (8-16m) are worth mentioning. The moisture content of this coal is low and the coals are medium quality coking coals.
- West Bokaro coalfield: The Barkar Formation is coal bearing and embodies 13 seams. The quality of the coal varies widely in this field. In most of the seams the ash content is high- greater than 25%. Most of the coals are medium coking coals.

3. KARANPURA COALFIELD

Just to the west of Bokaro coalfield lies the Karanpura coalfield. This coalfield is divided in to two parts – North Karanpura and South Karanpura . In Karanpura coalfield there is full development of Lower Gondwana and the younger rocks. In the North Karanpura a number of coal seams occur. The ash content of these coal seams is high. In South Karanpura there is good development of Talchir and Barkar Formation. Most of the coals of this area are non-coking.

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- ▶ North Koel Valley Coalfields
 - ▶ Deogarh Coalfields
 - ▶ Rajmahal Coalfields
 - ▶ Hazaribagh Coalfields

COALFIELDS OF CHHATTISGARH AND MADHYAPRADESH

(a) Central Indian coalfields

1. SINGRAULI COALFIELD

2. SOHAGPUR COALFIELD

3. UMARIA COALFIELD

(b) Satpura coalfields

1. MOHPANI COALFIELD

2. PATHAKHERA COALFIELD

3. KANHAN VALLEY COALFIELD

4. PENCH VALLEY COALFIELD

(c) North Chhattisgarh coalfields

(d) South Chhattisgarh coalfields (Hasdo- Arand, Korba, Mand-Raigarh)

COALFIELDS OF CHHATTISGARH AND MADHYAPRADESH

(c) North Chhattisgarh coalfields

1. TATAPANI RAMKOLA COALFIELD
2. JHILIMILI COALFIELD
3. SONHAT COALFIELD
4. CHIRIMIRI AREA
5. KURASIA AREA
6. BISRAPUR COALFIELD

(d) South Chhattisgarh coalfields

1. HASDEO – RAMPUR OR ARAND (SURGUJA) COALFIELD
2. KORBA COALFIELD
3. MAND-RAIGARH COALFIELDS

(e) COALFIELDS OF MAHARASHTRA

(i) Coalfields of Nagpur region

- 1. KAMPTEE COALFIELD**
- 2. BOKHARA COALFIELD**
- 3. UMRER COALFIELD**

(ii) Wardha Valley Coalfields:

(f) COALFIELDS OF ANDHRA PRADESH

Pranhita – Godavari Valley

(g) COALFIELDS OF ORISSA

Ib-River coalfields & Talcher coal fields